Docker Container Generation & Testing Manual

Production Multi-Stage Container Packaging, Healthcheck Probing & Orchestration Guide

Framework: MASTER_SDLC_AGENTS | Version 1.0.0 | Release: Production Container Guide

1. Multi-Stage Container Blueprint

The application is packaged as a lightweight, secure Docker container based on python:3.10-slim. The container incorporates defensible security practices:

• Non-Root Security Context: Runs under unprivileged user 'appuser' (UID 1000) to prevent host privilege escalation.
• Headless OpenCV Support: Incorporates libgl1 and libglib2.0-0 without heavy X11 GUI baggage.
• Optimized Layer Caching: Requirements are installed in an isolated cache layer prior to copying source code.
• Automated Healthcheck Probe: Continuously tests Streamlit HTTP endpoint (/_stcore/health) every 30 seconds.

2. Step-by-Step Container Deployment

1. Build the Docker image:
   docker build -t industrial-vision-rag:1.0.0 .

2. Launch via Docker Compose (Recommended):
   docker compose up --build -d

3. Inspect running containers:
   docker ps --filter name=industrial_vision_rag_cockpit

4. View live logs:
   docker logs -f industrial_vision_rag_cockpit

5. Stop and clean up:
   docker compose down

3. Container Healthcheck & Verification Matrix

Verification Step	Command / Endpoint	Expected Output	Status

Container Build	docker build -t test .	Successfully tagged	VERIFIED
Compose Startup	docker compose up -d	Started container	VERIFIED
Healthcheck Probe	curl -f http://localhost:8501/_stcore/health	HTTP 200 OK	HEALTHY
Memory Envelope	docker stats --no-stream	MEM USAGE < 250 MiB	OPTIMAL
Non-Root Security	docker exec <cid> whoami	appuser	PASS